the Ghatak achieves stealth while maintaining operational agility. The aircraft is expected to be powered by an indigenously developed Kaveri Dry Engine (non-afterburning variant), providing efficient thrust suitable for high-altitude, long-endurance missions (Ram, et al. 2022).

Technologically, the Ghatak integrates AI-driven mission control, autonomous navigation, and sensor fusion capabilities that allow it to perform target identification, route optimization, and coordinated strike missions with minimal ground intervention. Its internal weapons bay can house precision-guided munitions (PGMs), air-to-surface missiles, and electronic warfare pods, ensuring both stealth and offensive versatility. The autonomous mission management system enables the UAV to adapt dynamically to combat conditions, assess threats, and modify its mission trajectory in real time.

Operationally, the Ghatak is designed for strategic and tactical roles, including deep-strike penetration, suppression of enemy air defenses (SEAD), battlefield surveillance, and electronic intelligence (ELINT). Its integration with India's upcoming Advanced Medium Combat Aircraft (AMCA) program will enable manned–unmanned teaming (MUM-T), enhancing situational awareness and expanding India's aerial strike envelope.

The Ghatak UCAV signifies India's growing aerospace self-reliance and technological capability in developing indigenous stealth combat platforms. It combines stealth design, autonomy, and precision lethality in one system, marking a major leap in India's defense modernization efforts.

Figure 5.9 depicts the Ghatak UCAV (unmanned combat aerial vehicle), India's first indigenously developed stealth combat drone by DRDO–ADE. The image highlights its flying-wing design, optimized for low radar visibility through smooth, curved surfaces and radar-absorbent materials.